ACCELERATED MEDICAL IMAGING USING MACHINE LEARNING ARCHITECTURES

Multi-Modal AI Edge System | DICOM Integration | Bayesian Uncertainty Quantification | Continual Learning

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Hardware Target: NVIDIA Jetson Nano 4GB | Session 2022

EXECUTION: NVIDIA JETSON NANO

ORTHANC PACS SERVER
External simulated hospital node.
Sends/Receives DICOM SR.

◄ DICOM ►

ENTRY A: MODALITY
Real MRI/X-ray machine over DICOM Network → pynetdicom C-STORE

ENTRY B: MOBILE
Doctor's Smartphone / Laptop → HTTP POST → Flask REST API (:5000)

STAGE 0: OOD BOUNCER
Convolutional Autoencoder reconstructs input.
Computes Mean Squared Error (MSE) vs original.

REJECT: OOD
If MSE > Threshold: Log to SQLite, halt execution.

3-STAGE THREADED PIPELINE (PARALLELISM)

STAGE 1 (CPU)
- OpenCV byte decode
- Resize to 224×224 px
- ImageNet Normalization
- Reshape to [1,3,224,224]

Q#1 Buffer

STAGE 2 (GPU)
- CUDA memcpy H2D
- TensorRT FP16 engine
- Output: Logits + Features
- CUDA memcpy D2H

Q#2 Buffer

STAGE 3 (CPU)
- **MC Dropout:** 20x passes for Uncertainty%
- **GradCAM:** 1280 maps → JET overlay
- **Patient Fusion:** Late-Fusion MLP
- **Phi-2 LLM:** Generates clinical text via llama.cpp

CONFIDENCE-GATED ROUTING
Evaluates Output based on Mean Confidence & Std Deviation

TIER 1 (SAFE)
>90% Conf, <5% Uncert.
Auto-approve. Normal DICOM SR generated.

TIER 2 (WARN)
70-90% Conf OR 5-10% Uncert.
Flags DICOM SR for review.

TIER 3 (URGENT)
<70% Conf OR >10% Uncert.
Urgent review, pushes to Active Learning.

THE LEARNING LOOP (CONTINUAL & FEDERATED)

1. Active Queue
GUI displays Tier 3 cases. Doctor inputs true label.

2. EWC Continual
Loss = task + $\lambda\Sigma$(Fisher×$\Delta\theta^2$).
Hot-swaps TRT Engine.

3. Federated Sync
FedAvg across 3 simulated nodes. Distributes Global weights.

PHYSICAL I/O (GPIO)

- Tier 1 Auto-Approve (Pin 12)
- Tier 3 / OOD Alert (Pin 16)
- Piezo Buzzer Alert (Pin 18)

CLINICAL DASHBOARD UI

tab_1_scan.ui

```
[IMAGE VIEWER]
Overlay: GradCAM Heatmap (40%)
Diagnosis: Glioma Detected
Confidence: 94.2% ± 1.2%
```

tab_2_fusion.ui

```
[PATIENT RISK DASHBOARD]
History: MRI + ECG match found.
Fusion Score: CRITICAL RISK
```

tab_3_active.ui

```
[ACTIVE LEARNING QUEUE]
Pending Reviews: 4 cases
> Select Ground Truth: [DROPDOWN]
> [SUBMIT CORRECTION]
```

tab_4_tegrastats.sh

```
root@jetson:~# tegrastats
RAM 1845/3956MB (lfb 102x4MB)
CPU [34%,28%,15%,12%]
GR3D_FREQ 99%@921
POM_5V_IN 5120/5120
```

THE 6 MODALITIES

All utilize EfficientNet-B0 backbone

1. Brain MRI (Tumor/No Tumor → Subclass)
2. Chest X-ray (Pneumonia/Normal)
3. Retinal Fundus (DR Grade 0-4)
4. ECG Signal (Arrhythmia 1D→STFT)
5. Skin Lesion (HAM10000 7-class)
6. Bone X-ray (Fracture/Normal)